Artificial Intelligence Disclosures in Sustainability Reports: Towards an Artificial Intelligence Reporting Framework



Enrique Bonsón and Michaela Bednárová

Abstract With the rapid proliferation of the use of artificial intelligence (AI) in organizations over the last decade, certain concerns arise regarding human rights, data security, privacy or other ethical issues that could be at stake due to the uncontrolled use of AI. However, concerns regarding transparency in the use of AI are not yet reflected in any standards for the disclosure of non-financial information, nor in current regulations. Voluntary disclosure of AI, being a novelty, is scarce, implies a lack of standardization and is limited above all to the financial, technology and telecommunications sectors. Therefore, the main objective of this paper is to seek consensus and to propose a set of relevant elements to structure the information on the use of AI by companies, to improve transparency, mitigate risks and demonstrate a real responsibility in its use. For the purposes of this study, a set of disclosure elements had been proposed based on multi stakeholder approach with the collaboration between the New Technologies Commission of AECA and the BIDA Observatory. The final proposal has been validated by online questionnaires and includes a guide to the general information elements (AI governance model; Ethics and responsibility; Strategy) as well as more specific disclosure requirements for each medium–high risk automated decision making (ADM) systems. Thus, this research attempts to contextualize the development of artificial intelligence reporting standards.

Keywords Artificial intelligence · Disclosure · Standardization

E. Bonsón
University of Huelva, Plaza de La Merced 11, 21002 Huelva, Spain
e-mail: bonson@uhu.es

M. Bednárová (✉)
Pablo de Olavide University, Ctra. de Utrera, 1, 41013 Seville, Spain
e-mail: mbed@upo.es

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1 Introduction

The rapid evolution of the new technologies over the last decade has brought significant benefits to companies in terms of higher efficiency, cost savings, revenue generation, higher quality work, or increase employee satisfaction. Rapid technology adoption will only continue to accelerate as its increased efficiency becomes more evident. Indeed, intelligent automation is recently a key investment for businesses all around the globe. The major investments flow to business process management software, machine learning, deep learning, and robotic process automation [7].

Nevertheless, at the same time certain concerns are being raised regarding the human rights, data security, privacy or other ethical issues, which might be at stake when the AI is incorporated into decision making processes. In particular, disruptive artificial intelligence tools with a high level of automation, massive data collection and manipulation, and possible inherit biases are reasons for a wider society to worry. Thus, proliferation of these new applications increases both stakeholders' scrutiny and regulatory interest. Yet, despite its significant potential risks, recently there is no legislation put in place to mitigate the risks of AI usage by requiring companies to be fully transparent about it. Nevertheless, some companies have already started to respond to this request by voluntarily disclosing such information in their non-financial report. Inspired, in part, by European Commission initiatives such as the Ethical Guidelines for Trusted AI, published in 2019 [23], or the White Paper on AI: A European Approach to Excellence and Trust [24] that have laid the foundations for the European regulation of AI, Artificial Intelligence Act, published in April 2021 [25].

Over the last two decades we could have witnessed how organisations all around the world initiated and evolved the non-financial disclosure to complete their financial statements [8]. Initially, there was the same stakeholder pressure, which required companies to be transparent about their environmental, social and governance performance due to possible negative impacts on the environment and society as a whole. This further led to the development of generally accepted reporting standards such as Global Reporting Initiative (GRI), EMAS, ISO 26000, SA 80,000, etc., which developed guidelines and key performance indicators (KPIs) of the non-financial disclosure. Soon after, in some countries, the legislation was put in place and non-financial disclosure became obligatory, in particular, for large companies. The European Council, as one of the earliest promotors of non-financial reporting, on October 22, 2014, adopted a new Directive 2014/95/EU on the disclosure of Corporate Social Responsibility (CSR) by large companies and groups, which modifies the previously adopted Directive 2013/34/EU on annual financial statements, consolidated financial statements and related reports of certain types of companies [26]. This was followed by the publication of Commission Communication 2017/C 215/01 (Guidelines for the presentation of non-financial reports) [22].

In June 2020, the European Financial Reporting Advisory Group (EFRAG) have initiated work for the elaboration of possible EU non-financial reporting standards [27]. In line with the European Commission mandate, the project Task Force on

preparatory work for the elaboration of non-financial reporting standards has been carried out based on the multistakeholder approach within the European Corporate Reporting Lab. The main aim was to propose a roadmap for the development of a comprehensive set of EU sustainability reporting standards. The final report was published in February 2021 and provides 54 proposals for the elaboration of EU non-financial reporting standards. Their objective was mainly to describe the scope and structure of possible future sustainability reporting standards, not to set- out specific disclosure requirements, indicators or metrics. Built upon the proposed target standards architecture which would cover three layers of reporting (sector-agnostic, sector-specific and entity-specific), the ESG classification (the Planet, the People, and the Business) was recommended.

Regarding the social sub-topics, in the proposal #37, they point out that those might range from the Human Resources management perspective, to the broader Human Rights perspective. Thus, this disclosure should be aligned with international and EU reference frameworks and standards, including the UNGP on Business and Human Rights, the OECD Guidelines—and the other international declarations and principles—as well as with the Charter of Fundamental Rights of the EU. Nevertheless, a clear articulation of what should be reported in this category remains to be further explored and challenged. Similarly, despite that there are no specific instructions on how to report on data privacy, this issue has been raised within the proposal #38, related to Governance sub-topics, in particular, related to business ethics. Although there is no specific mention regarding the AI disclosure, due to its strong relation with the human rights and possible infringements of them, this area is worth exploring as a possible future component of the sustainability disclosure.

On 21st of April 2021, the European Commission proposed a legal framework, which serves as a proposal for a regulation on a European approach for AI, together with a coordinated plan with Member States on AI [25]. The main aim is to define rules and actions for excellence and trust in AI within the scope of a Europe fit for the Digital Age and to guarantee the safety and fundamental rights of people and businesses. The framework follows a risk-based approach and distinguishes four categories of AI systems: (1) unacceptable risk; (2) high-risk; (3) limited risk; and (4) minimal risk. Depending on the level of risk, the company would be obliged (or not) to a certain level of disclosure. E.g., the AI systems classified in a category of unacceptable-risk (applications and AI systems that could cause physical/psychological harm, manipulate human behaviour or exploit human vulnerabilities, or any application that would allow social scoring or real time remote biometric identification in public places) will automatically be banned. High-risk AI systems would require a comprehensive risk management and to guarantee quality of data, accuracy, robustness, traceability of results and cybersecurity. Limited-risk AI systems will be subject to lower transparency requirements (e.g., users should be at least aware that they are interacting with an algorithm). On the other hand, for minimal-risk AI systems no disclosure will be required (e.g., video games or spam filters).

Reporting on AI, being quite a novelty, implies a lack of standardisation in such a disclosure. To standardise it, first, a consensus has to be reached on what and

how to report; what would be the key elements and what information should be disclosed to satisfy the information needs of different stakeholders. Therefore, the main objective of this paper is to propose and validate a set of general elements that should be disclosed to structure the information about AI (an important part of corporate digital responsibility—CDR); improve the transparency; mitigate the risks and prove a real responsibility in the usage of AI.

The paper can be considered one of the first attempts to structure and standardize the information related to AI in the non-financial disclosure context. Thus, a proposal and an external validation of the AI reporting framework can contribute to both literature and practice. Practical implications of this paper lays in helping companies to organize the information related to AI in a structured way focusing on relevancy. This research can be used as a springboard for further discussion, analysis and standardization.

2 Background

2.1 AI Disclosure Practices in European Companies

Nowadays, only few companies report on AI in their sustainability or annual reports. The early adopters of this new form of non-financial disclosure are mostly companies operating in financial, technological and telecommunication sector.

Recent study on AI ethical disclosures in corporate reports from 2018 and 2019 analysing IBEX 35 and DAX 30 companies shows that this type of disclosure is in its early stage and the companies providing such a disclosure are mostly from technological and telecommunication sector [9]. Similarly, a study regarding the AI disclosure in the annual reports of IBEX-35 companies show that this kind of disclosure is in a very early stage [10]. Nevertheless, the recent initiatives such as Ethics Guidelines for Trustworthy AI issued by the European Comission (2019) as well as the White paper on AI [24] are expected to increase the interest in AI transparency in the upcoming years.

Another recent study on ADM disclosure on the sample consisted of 962 annual/sustainability reports of the listed companies on the Western Europe countries (13 countries) stock markets published in 2018 and 2019 shows that ADM disclosure has still a long way to go and the first adopters are mostly companies operating in financial sector. Only 20 Western Europe companies disclosed the information related to ADMs in their annual/sustainability reports. Main categories of ADM disclosure that were identified in our study were: ADM for credit risk assessment (CRA); ADM responsibility; medical algorithms and diagnostic; and other. The latter being related to electronic sector disclosing the AI technology used in devices such as smartphones, smartwatches or home robots. The most frequent disclosure was related to CRA by companies operating in banking and financial services and ADM responsibility.

Nevertheless, despite the fact that the ADM for CRA was the most frequent reporting category, disclosure itself was rather general and there was a lack of details on which processes are fully automated and which are just assisted by the AI.

The second most frequently reported category was ADM responsibility. However, the way how this category was reported on shows weaknesses as well. We might have observed different approaches in this type of disclosure. Some companies were simply expressing general concerns regarding unacceptable effects of the decisions made by AI or regarding future scenario when the decisions will be taken only on the basis of some algorithm. Other companies were more specific regarding the ADM responsibility toward the issues such as avoidance of bias and discrimination; digital rights; or transparency that ADM is being used for personal data processing and clients' profiling.

However, extensive research is needed to properly explore what are the current practices in AI reporting of large European companies, and what are the determinant factors and incentives for such a disclosure.

2.2 Previous Studies

The studies of Mason [44] and Moor [45] were one of the first studies dealing with the information technologies' responsibility followed by studies focusing on ethics in emerging IT [13, 14, 17, 29, 51].

When the AI has become an integral part of business processes, concerns related to AI ethics started to be a topic for investigation [4, 9, 16, 28, 29, 33, 34, 36, 37, 50, 58] as well as data protection and privacy [3, 5, 12, 19, 41].

The term corporate digital responsibility is relatively new but there already are some studies focusing on this topic [39, 42, 43]. Nevertheless, the scope of those studies is rather narrow and there is a lack of conceptualization. On the other hand, there is an increased attention paid to this area by professionals.

2.3 Corporate Digital Responsibility

The Institute of Consumer Policy provides the following definition of CDR: "*companies' responsibility for the consequences of their business processes, products and services for employees, suppliers, customers, society as a whole and the environment encompassing: (1) data and algorithmic decision making, (2) participation and reduction of inequality, (3) digital education, (4) future of work, and (5) digitalization in service of an ecologic transformation*" [18].

According to Lobschat et al. [43] who developed a framework of CDR culture, organizations should be transparent about their responsibility in the digital age. Thus, CDR should focus on the ethical aspects that are inherent to the digital context.

2.4 *Corporate Digital Responsibility as a New Layer of Corporate Social Responsibility*

Although CDR has its own particularities due to the digital context [43, 54], some authors see a clear connection between the CDR and CSR as they share the same basic principles related to environmental and social concerns [55]. Early adopters organizations consider CDR a complementary aspect of CSR [21, 31, 48]. Indeed, professionals [21, 48] point out the importance of CDR to increase stakeholders' trust in the digital world and call for extending the scope of CSR by CDR arguing that companies have embraced the environmental and social responsibility idea but the digital revolution has brought new responsibilities. Gärtner et al. [31] point out the importance to adjust the way in which companies approach their responsibility as the digitalization affects almost all aspects of our lives.

2.5 *Theoretical Background (Legitimacy Theory)*

As CDR can be considered a new layer of CSR [21, 31, 48], the question arises, how the motivation of organizations to voluntarily disclose information about AI systems could be explained.

CDR approach can increase stakeholders' trust in digitalization [48]. As several studies suggest that legitimacy theory may provide a satisfactory explanation of voluntary non-financial reporting in general [6, 15, 20, 32, 35, 46, 53]; considering the parallel between the two, legitimacy theory seems to offer a reasonable explanation for voluntary AI or ADM disclosure [10].

Corporate legitimacy is a general assumption that the actions of a company meet the norms, values and expectations of a society [56]. Companies need to legitimate their practices and gain the acceptance from stakeholders in order to conduct their business successfully [20]. Higher transparency helps to increase the trust to the company and therefore improves its legitimacy. To gain the legitimacy, companies should continuously adjust their disclosure practices by focusing on the most relevant issues. By other words, they should focus on topics that raise certain concerns of a wider society in a particular moment of time. Nowadays, such concerns are related to the increased use of disruptive AI systems, in particular those that represents high risks and might raise some ethical issues such as facial recognition or black box ADMs [47, 57].

2.6 *Technology Overview*

Before opening a discussion about which information elements should be included into AI reporting framework, a brief technology overview is provided.

Intelligent automation refers to the use of technology to conduct work task, those are i.e.: Robotic desktop automation (RDA); Robotic process automation (RPA); Visual recognition technology; Artificial Intelligence (AI). AI helps automation evolve by combining Machine Learning (ML), predictive and adaptive models, natural language processing (NLP) and Deep Learning (DL) as a subset of ML.

2.6.1 AI Versus Automation

While automation focuses on streaming repetitive tasks, saves time and cost on monotonous and voluminous tasks, AI mimics human intelligence decisions and actions. Therefore, AI technologies and processes mimic how human beings react, speak, hear (even understand), and make decision. Machine Learning (ML) as a subset of AI, learns from data, identifies patterns and recommend decisions without human involvement. Deep Learning (DL) as a subset of ML, is based on artificial neural networks which continuously learns and even adjusts itself automatically to improve accuracy.

There is an immense variety of AI tools. Therefore, it is important to distinguish AI that is autonomous in making decisions and learns from data on its own, but the purpose and scope of its usage does not imply any harm or ethical issues and those, which are highly autonomous, learn and train themselves and might imply more harm to individuals by area where they are applied.

The first group would be AI applications such as textio, AI tool using predictive technology to write job listings to make them sound lucrative to potential candidates. Companies such as Twitter, Microsoft, Starbucks are using it. Another example of so called "low risk" AI app would be conversica, which is used to manage gross sales needs. It is an AI app that connects with the leads through automated two-way email conversation. X.ai is a smart assistant for handling meetings; Digital Genius is based on NLP and is able to maintain human like conversations with customers. Intraspexion is an AI tool to prevent and warn companies against litigation risks by regularly scanning company's internal emails and flagging potential litigation risks for future review. Similarly, a facial recognition app which is used for tagging photos on the social network such as FB, does not imply any harm.

3 Material and Methods

A set of indicators was proposed based on the European Commission's White Paper on AI—A European approach to excellence and trust (2020), Artificial Intelligence Act (2021) and Digital Rights Charter—XXIII Digital rights related to AI [52], and discussions with the BIDA Observatory and the New Technologies and Accounting Commission of AECA. The collaboration between BIDA Observatory and New Technologies and Accounting Commission of AECA represents a combination of business world, public organizations, regulatory bodies and academics.

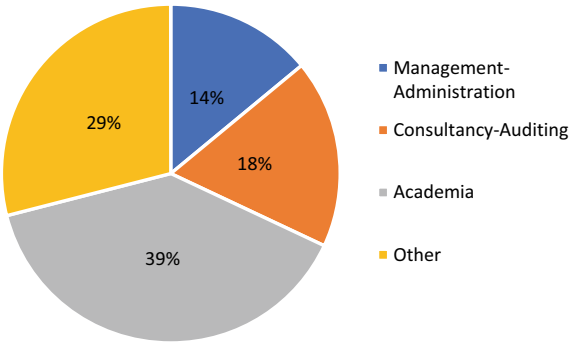


Fig. 1 Respondents based on professional profile

Hence, following a multistakeholder approach. Such a proposal was later validated by international respondents through online questionnaire.

The objective of the questionnaire (Appendix 1) was to know the opinion about the relevance of each of the elements proposed as a guide to disclosure of information related to the use of AI in the non-financial information statement. The respondents were asked to rate each item on the scale 1 (least relevant) to 5 (most relevant) based on how important they think the information might be in the non-financial reporting context. They were also asked to add any other item they would consider relevant.

The final proposal of the elements was approved by the members of the New Technologies and Accounting Commission of AECA and the BIDA Observatory through followed-up discussions. The questionnaires were sent in the period of January-March 2021. In total, 44 questionnaires were responded (response rate 58.6%). Professional profile of the respondents is depicted in Fig. 1.

4 Results

Based on the scale 1 (least relevant) to 5 (most relevant), the average evaluation of proposed reporting items is depicted in the Table 1. As can be seen, every item received at least 4 points on average, which implies a high relevance of a proposed particular item in the context of non-financial reporting.

After the validation of the proposed reporting elements (all of the items are considered validated as they had obtained equal or more than 4 points on the scale 1–5) a framework which consists of two main parts: general elements and specific ADM disclosure was developed.

Table 1 Average evaluation of proposed reporting item

Category	Reported item	Evaluation
AI governance model	GM1	4.3
	GM2	4.4
	GM3	4.2
Ethics and responsibility	ER1	4.4
	ER2	4.3
	ER3	4.0
Strategy	S1	4.4
	S2	4.1
	S3	4.1
	S4	4.1
ADM disclosure	ADM1	4.4
	ADM2	4.2
	ADM3	4.3

4.1 General Elements Disclosure

When disclosing the information about AI, it is important that each organisation first describes the context in which the AI technology is used, why and how. As AI disclosure is rather a new trend, there is a lack of guidelines on what and how to report. Therefore, below we provide a brief guidance on narratives related to general elements, which we consider relevant to be reported on.

Nevertheless, there is an urgent need for internationally recognized and generally accepted AI reporting Standards, which would represent global consensus and best practice for reporting publicly the impacts of organisation's AI usage. AI reporting based on the Standards would also aim to guarantee an organization's commitment to fundamental rights such as freedoms, privacy, data protection and non-discrimination while incorporating AI technology, by other words it would demonstrate its corporate digital responsibility (Table 2).

I. *Governance Model and System Control*

Organisations should be transparent about how the AI is governed within the organisation. Hence, the following information should be provided: what is the governance structure; who is being held for responsible; existence of departments or committees such as data protection officer or monitoring committee; information about continuous monitoring as well as external verification systems and processes.

II. *Ethics and Integrity*

Some companies might rely on international principles regarding the ethical use of AI such as IEEE, EU, OECD, etc., while others might design their own principles or code of ethics related to the AI usage. In this point, the organisation should provide some information about the main ethical issues as well as mechanism for advice and

Table 2 Information about AI in the non-financial report

General elements	
I. AI governance model	(GM1) Governance structure, roles, responsibilities, departments and committees
	(GM2) Continuous monitoring, control and internal verification
	(GM3) External verification systems or processes (of black box ADM and high sensitive applications)
II. Ethics and responsibility	(ER1) AI principles followed by the company (own, EU, IEEE, OECD, etc.)
	(ER2) Relevant ethical issues, especially in relation to sensitive applications such as facial recognition, black box ADMs and recommendation systems
	(ER3) Mechanisms for users to receive meaningful explanations
III. Strategy	(S1) AI risk and impact analysis
	(S2) Relations with stakeholders, participation in organizations and forums related to the responsible development of AI
	(S3) Training activities (internal and external) on responsible use of AI
	(S4) Projects focused on meeting the Sustainable Development Goals (SDG)

concerns about ethics regarding the responsible use of AI. Thus, the scope of AI disclosure should be outlined starting with the list of material topics such as privacy matters; the usage of highly sensitive apps such as facial recognition; and ADM.

In addition, there should be a contact point where individuals can direct their queries regarding AI report as well as information about the fundamental rights of individuals who were treated by the AI of the company.

III. *Strategy*

Perhaps the most relevant part in this section would be how the AI related risks are identified and managed. Company can also report on ongoing training activities related to responsible usage of AI and any projects focused on SDG, where the AI plays a crucial role. Relations with stakeholders, participation in organizations and forums related to the responsible development of AI can also be disclosed here.

Nevertheless, the most attention should be paid to the risk assessment. Some applications such as the facial recognition might be more sensitive for a societal approval due to various practical or ethical implications. To identify which applications implies higher risks than other, a proper risk assessment has to be conducted and described.

Company should be transparent about in which areas of the company the AI is being involved. Some applications might involve more ethical issues due to the purpose they are used for, e.g., AI used in the HR, which involves decision about humans in comparison to AI in the manufacturing. Other applications become riskier when they are left autonomous with low or no human supervision. Thus, the purpose

of AI usage and the level of automation and independence should be considered as relevant factors in the risk assessment.

4.2 ADM Transparency

4.2.1 Risk Assessment Following the EC Approach

After performing the risk assessment, AI applications should be categorized based on their risk level. This way, applications with higher risk level would require more detailed disclosure on how the data are collected, stored, and managed; how the algorithm works; potential risks, biases and mitigation plans; strategy to ensure transparency, auditability, explicability, accessibility, usability, trust; and verification internal/external.

More transparency, for example, would be required for black box ADM in the critical areas as they imply the highest potential risk.

The aim of distinguishing 4 level AI risk categories, such as the approach of EC, is twofold, to protect individuals against negative impacts of highly sensitive AI applications, and not cause unnecessary burden for companies which are not using invasive AI technology.

4.2.2 ADM Disclosure

As described above, a special attention should be paid to highly autonomous AI tools—algorithmic decision systems (ADS) also known as Algorithmic decision making (ADM) which are based on the analysis of large amount of data to infer correlations or to derive information to make decision. Such applications imply, per se, potential practical and ethical concerns as the decision making is left on the machines. Humans are error-prone and biased, but systems can be biased too. That is why, it is important to know who built them, how they are developed and how they are ultimately used. To avoid bias in the algorithm, it should be fed with a representative dataset, the right model should be chosen, and the algorithm should be continuously reviewed and monitored. Therefore, the transparency on these aspects might not only increase the trust to the ADM but also serve as a proof of a company's commitment to corporate digital responsibility (CDR).

Regarding the ADM, we should distinguish between those which are rather predictable, so with certain input we obtain an expected and repetitive output (such as ADM in automating quality control) and those which are based on Machine Learning (so the decision is made within a "black box"), in particular, when they might have an impact on individual's life. ADM processes are fed with data, which might be biased. If there is a bias in the data the ADM is trained with, this bias would be reinforced and amplified which ultimately leads to unfair or discriminatory decision.

Table 3 ADM disclosure framework

ADM disclosure	
Automatic decision-making system (ADM) transparency (of systems that may imply a potential risk for the subject on whom it is decided)	(ADM1) System description: type, input data, algorithm design, human supervision, etc
	(ADM2) Statement that the system complies with the digital rights regulation related to AI or, if applicable, that algorithmic non-discrimination, transparency, auditability and explicability of the algorithm are ensured, as well as its accessibility, usability and reliability
	(ADM3) Verification of compliance by independent agents

Thus, the results yield by the ADM can be discriminatory without decision-makers to be motivated to discriminate. E.g., there might be some variables, such as gender or race, which normally can not be taken for consideration while making a decision but are often statistically associated with seemingly inoffensive characteristics, such as height or postal code. As ADM works with huge datasets of correlated data, it can lead to indirect discrimination. Therefore, individuals should have two basic rights when the decision has been made based on the "black box" ADM, (1) right to a meaningful explanation and (2) right to contest. Hence, companies should ensure individual awareness of the processes where "black box" ADM has taken place, as well as of their associated rights.

Thus, the disclosure in this section should cover basic system descriptions such as: type of algorithm, input data, algorithm design, information if there is human supervision or not, etc. Together with the statement that the system complies with the digital rights regulation related to AI or, if applicable, that algorithmic non-discrimination, transparency, auditability and explicability of the algorithm are ensured, as well as its accessibility, usability and reliability. The company can increase the credibility of such a disclosure by having it verified by independent agents (Table 3).

5 Discussion

As CDR seems to be a new layer of CSR, we might expect certain changes in corporate disclosure in the upcoming years [31]. Following the pattern of CSR disclosure evolution, first we might expect early adopters, normally large corporations, which will voluntarily disclose CDR aspects in order to legitimate their digital practices. This will likely continue with seeking a global consensus on what and how to disclose which would be later followed by legislation [7]. Nevertheless, while CSR disclosure took more than two decades until the legislation was put in place, with CDR we face higher urgency to address the risks associated due to the exponential advancements

of technology where the damage might take decades to repair, in particular, when we are talking about the ADM.

The aim of this paper is to draw attention to an emerging trend, AI usage and reporting, discuss the need of a global Standard for such a disclosure and contribute to its creation by proposing the draft of general elements, which should be reported on.

The main aim of the new Standards for AI disclosure should not be to cause more burden on the companies to prepare their non-financial reports or to demonize the usage of AI which can be indeed highly beneficial, but there are certain fundamental rights of individuals that should be guaranteed. Hence, it is important to act quickly as the negative impacts of irresponsible AI usage might take decades to correct.

The main concerns are human rights, data security, privacy or other ethical issues that could be at stake due to the uncontrolled use of AI. Therefore, on one hand to require from companies to be more transparent about the use of AI and on the other hand, to provide them with a clear guidance on what and how to report might help society as a whole to prevent undesirable events to happen. For better illustration, we can focus on e.g., privacy, a fundamental human right, some AI are based on personal information which has consequences for their privacy. Initiatives such as Personal Information Protection and Electronic Documents Act (PIPEDA) in Canada but also institutions such as EU point out the importance of privacy as a fundamental human right, therefore, when organizations want to feed their AI tools with personal data, they should complete so called privacy impact assessment (required in Canada) form and perform balancing test in which they compare the necessity and proportionality of the measure with interests, fundamental rights and freedoms of individuals. Such transparency and accountability should help to avoid collecting data for data sake (or even doing business with data) and to use disproportional measures (e.g. the usage of facial recognition if other less invasive method could be applied based on less sensitive data). Hence, the main aim of this paper is to contribute to a creation of such an accountability tool, which would deal with this and other ethical issues that might arise.

6 Conclusions

Given that the AI technology in a modern and interconnected world is a global problem, global consensus based on multistakeholder approach should be pursuit. Only this way, we as a society can guarantee the respect of fundamental rights such as freedoms, privacy, data protection and non-discrimination while using AI technology and benefiting from it.

This study is one of the first attempts to standardize AI disclosure by proposing and validating the most relevant elements of AI reporting and developing an initial AI reporting framework. Nevertheless, before closing, limitations of the study should be acknowledged and future research outlined. Still, extensive research is needed to

properly explore what are the current practices in AI reporting of companies, and what are the determinant factors and incentives for such a disclosure.

The proposal of the elements was validated through multistakeholder approach by the members of the New Technologies and Accounting Commission of AECA and the BIDA Observatory and may represent a first attempt toward the AI disclosure standardization. Nevertheless, due to a fast evolution of AI technology, a continuous assessment of disclosure elements should be performed.

Appendix 1

QUESTIONNAIRE: Information About AI in the Non-financial Report

The objective of this questionnaire is to know your opinion about the relevance of each of the elements that are proposed as a guide to disclosure of information related to the use of Artificial Intelligence (AI) in the non-financial information statement.

Please rate each item based on how important you think it might be in the non-financial reporting context. Use the scale 1 (least relevant) to 5 (most relevant).

Your professional profile*

- Consultancy-Auditing
- Management-Administration
- Academia
- Other:

Company or institution*

Your answer

AI governance model

Governance structure, roles, responsibilities, departments and committees.

Continuous monitoring, control and internal verification systems.

External verification systems or processes.

Ethics and responsibility

AI principles followed by the company (Own, EU, IEEE, OECD, etc.).

Relevant ethical issues, especially in relation to sensitive applications such as facial recognition, black box ADMs and recommendation systems.

Mechanisms for users to receive meaningful explanations.

Strategy

AI risk and impact analysis.

Relations with stakeholders, participation in organizations and forums related to the responsible development of AI.

Training activities (internal and external) on responsible use of AI.

Projects focused on meeting the sustainable development goals.

For each automatic decision-making system (ADM) that may imply a potential risk for the subject on whom it is decided:

System description: type, input data, algorithm design, human supervision, etc.

Statement that the system complies with the digital rights regulation related to AI or, if applicable, that algorithmic non-discrimination, transparency, auditability and explicability of the algorithm are ensured, as well as its accessibility, usability and reliability.

Verification of compliance by independent agents.

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